MIL-P-60504A (MU)
18 June 1970
SUPERSEDING
MIL-P-60504 (MU)
29 September 1966

MILITARY SPECIFICATION

PRINTED CIRCUIT BOARD ASSEMBLY - Q-SWITCH TIMING CIRCUIT: 10541270

1. SCOPE

1.1 This specification covers a solid state control assembly, which is a modular component of the Range Finder AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical, for Coating Printed Circuit Assemblies

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

FSC 1240

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DRAWINGS

U.S. Army, Frankford Arsenal

F10540501	Range Finder, Laser, AN/GVS-3
C10540647	Connector
C10540938	Connector
F10540966	Cover
C10541003	Connector
F10541270	Printed Circuit Board Assembly-Q-Switch Timing Circuit

PACKAGING DATA SHEET

10541270	Packaging of Printed Circuit Board Assembly - Q-Switch Timing Circuit

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The printed circuit board assembly - Q-switch timing circuit, hereinafter referred to as assembly, shall be manufactured in accordance with Drawing F10541270 and drawings pertaining thereto.

3.3 General specification.- The assembly shall meet the following requirements, where applicable, of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926, the assembly shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.11.

3.3.1.1 Etched circuit board.- The conductor pattern on the etched board shall be free from blisters, wrinkles, cracks, holes, dents, scratches, and all other defects that will affect serviceability. The pattern shall be free of loosely adhering feather edges of metal. In no case

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shall a finished line width be more than 0.010 inch narrower than that specified on the printed circuit master. In no case shall a conductor line width be less than 0.020 inch.

3.3.1.2 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with at least one point of the component in contact with the board. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface.

3.3.1.3 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that the proper identification markings of each part is visible.

3.3.1.4 Transistors and integrated circuits.- Transistors and integrated circuits are to be mounted perpendicular to the circuit board, with spacers between the component and the circuit board.

3.3.1.5 Bend radius.- The minimum bend radii for formed leads shall be one lead diameter. The maximum shall be two diameters or 0.062 inch, whichever, is smaller.

3.3.1.6 Component leads.- All leads are to be covered with sleeving so that no more than 1/32 inch bare lead is exposed. Leads of transistors mounted with a spacer directly into a printed circuit board, shall have 1/16 inch of lead from the transistor base to any soldered joint. Transistors mounted with a spacer require no sleeving on the leads. Component leads shall be preformed to suit mounting holes, there shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than 3/64 inch, including solder, beyond the side of the circuit board, opposite the component.

3.3.1.7 Plated through connections.- All plated through connections shall be filled with solder in order to eliminate dependence on plating itself for circuit continutiy.

3.3.1.8 Heat sensitive component.- The use of heat shunts or cooling devices shall be mandatory when hand soldering components that are subject to damage by heat.

3.3.1.9 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 pounds per square inch (psi). The temperature of the cleaning solution shall not exceed 125°F. After cleaning, there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

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3.3.1.10 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than the room temperature drying method is used, the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements:

a. Permissible dust particle size not to exceed 6 microns
b. Permissible moisture content limited to 2.12 grains per cubic foot (dew point 32°F maximum)
c. Dust count shall not exceed 1000 particles per cubic foot

3.3.1.11 Coating.

3.3.1.11.1 Masking.- Prior to coating with insulating compound MIL-I-46058 the following areas shall be masked; adjustment screw of R22 Resistor, the R12, Resistor areas on each side of board around the 6 mounting holes, areas of the interlock insulator, Connectors J1 and J2 and the ten test points. Subsequent to coating, with masking removed all above areas shall be clean and free of coating.

3.3.1.11.2 Coating, insulating compound.- The assembly shall be coated with an insulating compound. The insulating compound shall be Insulating Compound, Electrical, Type PUR, Polyurethane Resin conforming to MIL-I-46058.

3.3.1.11.3 Cured compound.- The cured compound shall represent a smooth continuous surface, and shall pass from the mounted components to the circuit board with no breaks at their junction. Except for areas masked, all surfaces shall be covered with a coating of 0.001 to 0.006 inch thickness. The cured compound shall meet the following requirements:

a. Compound shall be free of visible dust particles or foreign matter.
b. It shall be smooth and uniform and have no globules or bare spots.
c. It shall be substantially free of bubbles.
d. There shall be no blisters or cracks.
e. It shall be transparent and shall not affect marking legibility.
f. It shall be capable of being removed in local areas for replacement of components by means of a soldering iron or similar method.

g. Applied to a test board of 1/16 inch thickness, the compound shall not rupture or separate from the board, when the board is bent into an arc of twelve-inches radius.

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3.4 Environmental.

3.4.1 Storage temperatures.- The assembly shall withstand thermal stabilization at temperatures of -80°Fahrenheit (F) and +160°F. Subsequent to temperature exposure, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.4.2, 3.6 and 3.7.

3.4.2 Operational temperatures.- The assembly shall meet the requirements of 3.5 through 3.5.8 when thermally stabilized at temperatures of -65°F, +155°F and at room temperature of between +70° and +80°F.

3.4.3 Shock.- The assembly shall withstand a total of 18 shock impulses, three in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75 g's. Subsequent to shock, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.4.2, 3.6 and 3.7.

3.4.4 Vibration.- The assembly shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second. The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5 g level at higher frequencies. Subsequent to vibration the assembly shall show no evidence of physical damage and shall meet the requirements of 3.4.2, 3.6 and 3.7.

3.5 Performance.- The assembly shall meet the requirements of 3.5.1 through 3.5.8 for input voltages (Ei) of 21.0 ± 0.1, 26.5 ± 0.1, and 30.0 ± 0.1 VDC under conditions shown in Figure 1. Each input voltage shall be applied continuously, except as specified in 3.5.8.

3.5.1 Input current (quiescent state).- The current drawn by the assembly shall not exceed 10 milliamperes before operation is initiated.

3.5.2 Operation initiation.- Operation of the assembly shall be initiated by the application of the pulse described in Figure 2 to Pin 1.

3.5.3 Input current (operating state).- The current drawn by the assembly during steady state operation shall not exceed 350 milliamperes.

3.5.4 Regulated output voltage.- The assembly shall supply a regulated 17.5 ± 0.25 volts across a load between Pins 3 and 7 as shown in Figure 1. The assembly shall be capable of supplying an initial surge current to load of at least 5 amperes. The sustained current to load shall be 300 ± 10 ma.

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3.5.5 Internal reference.- The monostable multivibrator which generates the internal reference shall be adjustable to a setting of 2500 microsecond pulsewidth by R22 and shall not vary from the initial setting by more than 25 microseconds.

3.5.6 Voltage control.- The assembly shall compare the frequency of a square wave input applied to Pin 15 with an internal reference and supply the voltage of 3.5.4 while the frequency of the input signal is less than that of the internal reference. When the frequency of the applied signal equals that of the internal reference, the output of 3.5.4 shall no longer be present. Once removed, this output shall not appear until operation is again initiated by 3.5.2.

3.5.7 Pulse outputs.- The following pulses shall be generated when the frequency of the applied signal equals that of the internal reference:

a. A positive pulse of 1.5 ± 0.25V amplitude and 750 ± 50 microsecond duration (see Figure 3) shall be developed at J2 across a 200 ohm load.

b. A negative pulse of 9.25 ± 0.25V amplitude and at least 50 microsecond duration (see Figure 4) shall be developed at Pin 13 of J1 across a 150 ohm load.

c. A differentiated output pulse of 5 ± .25V, amplitude and 70 ± 10 microsecond duration (see Figure 5) shall be developed at Pin 11 of J1 across a 6.8K ohm load.

3.5.8 Power off - pulse output.- A positive pulse of at least 1.5V in amplitude and 25 microsecond duration shall be generated at J2 when the input voltage is removed from the assembly.

3.6 Electrical continuity.- The electrical continuity of the assembly shall conform to the electrical Schematic Drawing F10540935.

3.7 Interface requirements.- The assembly shall interface with:

a. Mounting pads and holes of Cover 10540966.

b. Connector 10540647 (13P2).

c. Connector 10541003 (23P1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect

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(individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0% DEFECTIVE			
101.	Electronic workmanship	3.3.1	Visual
102.	Etched circuit board	3.3.1.1	SME; Visual
103.	Spacing of parts	3.3.1.2	SME; Visual
104.	Symbols and markings	3.3.1.3	Visual
105.	Transistors and integrated circuits	3.3.1.4	Visual
106.	Bend radius	3.3.1.5	SME; Visual
107.	Components leads	3.3.1.6	SME; Visual
108.	Plated through connections	3.3.1.7	Visual
109.	Heat sensitive components	3.3.1.8	Visual
110.	Masking	3.3.1.11.1	Visual
111.	Cured compound	3.3.1.11.3	SME; Visual
MINOR: NONE			

NOTE: The above characteristics shall be inspected at in-process intermediate inspection stations.

TABLE II - CLASSIFICATION OF DEFECTS

(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
112.	Operation initiation	3.5.2	4.6.2.2
113.	Regulated output voltage	3.5.4	4.6.2.2
114.	Voltage control	3.5.6	4.6.2.3
115.	Pulse outputs	3.5.7	4.6.2.4
116.	Input current	3.5.1, 3.5.3	4.6.2.5
117.	Internal reference	3.5.5	4.6.2.2
118.	Power off pulse	3.5.8	4.6.2.6
119.	Electrical continuity	3.6	4.6.3
120.	Interface requirements	3.7	4.6.4
MINOR: NONE			

NOTE: The tests in Table II shall be conducted at a temperature between +70° and +80°F.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.1.9
Drying	3.3.1.10
Coating, insulating compound	3.3.1.11.2

4.4 Special sampling.

4.4.1 General.- Three assemblies shall be selected at random as a special sampling from each 50 produced. The samples shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable Dwgs & Specs - Visual
302.	Fabrication	3.2	Applicable Dwgs & Specs - Visual
303.	General specifications	3.3	MIL-F-13926 - Visual

NOTE: The tests in Table III shall be conducted at a temperature between +70° and +80°F.

4.4.2 Environmental.- Three assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever, occurs first. Each sample shall be subjected to the tests in Table IV and shall then meet the requirements and tests of Table II.

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TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperatures (-80°F and +160°F)	3.4.1	4.6.1.1
305.	Operational temperatures (-65°F and +155°F)	3.4.2	4.6.1.2
306.	Shock - between +70°F and +80°F	3.4.3	4.6.1.3
307.	Vibration - between +70°F and +80°F	3.4.4	4.6.1.4

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests specified in 4.6:

	Nomenclature	Description
1.	D.C. Power Source	Range of 0-30 volts, current rating 5 Amps, impedance less than 0.1 ohms.
2.	Square Wave Generator	Capable of providing a 1.5 V amplitude square wave of frequencies from 200 to 500 hertz.
3.	Time Interval Counter	Accuracy of ± 1 count, ± 2 micro-seconds, input impedance 100 kilo ohms, 20 picofarads or less.
4.	Ammeter	Capable of withstanding a 5 amp surge current and measuring 300 milliamperes within 1% accuracy.
5.	Voltmeter D.C.	Range 0-50 volts, accuracy 1%; sensitivity of 20 ,000 ohms volt.
6.	Oscilloscope	Min. requirements: rise time 20 nano-seconds, input impedance of 10 megohms, time base of 0.1 microseconds, to 20 milliseconds per centimeter, sensitivity of 0.1 volt per centimeter.

4.6 Test methods and procedures.

4.6.1 Environmental.- The assembly shall be thoroughly examined for physical defects after each environment test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the assembly shall be subjected to performance tests specified in 4.6.2 through 4.6.4.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperatures specified in 3.4.1.

4.6.1.2 Operational temperatures.- The test shall be performed in accordance with Procedure II of MIL-F-13926 at the extreme temperatures of -65° and +155°F utilizing the test equipment indicated in 4.6.1.1. The assembly shall be exposed to each specified temperature for a minimum period of 2 hours. At the end of each 2 hour period, and while still at the specified temperature, the assembly shall be tested in accordance with 4.6.2 through 4.6.2.6.

4.6.1.3 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

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The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the waveform, time duration and peak "G" value specified in 3.4.3.

4.6.1.4 Vibration.- The assembly shall be positioned as specified in 3.4.4 and firmly mounted to the vibration tester. The assembly shall be subjected to Test Method 514, Procedure I, Part I of MIL-STD-810, except that the acceleration, amplitude, frequency, and time duration shall be as specified in 3.4.4.

4.6.2 Performance.- Prepare test set-up as shown in Figure 1. This set-up shall be maintained for all performance tests. The D.C. power source (Item 1 of 4.5.3) shall be connected between Pin 9 and ground with Pin 9 connected to the positive side of the power source. Performance tests indicated in 4.6.2.2 through 4.6.2.6 shall be performed at each input voltage (Ei) as specified in 3.5.

4.6.2.1 Pre-operation adjustment (70° to 80°F).- Apply an input voltage (Ei) of 26.5 ± 0.1 volts D.C. to Pin 9. Insert a 300 hertz square wave from square wave generator (Item 2 of 4.5.3) to Pin 15. Apply the input waveform as specified in Figure 3-1 to Pin 1. Utilizing a time interval counter (Item 3 of 4.5.3), measure the time duration of the lower voltage level of the waveform appearing at test point TP2. Adjust R22 to establish a 2500 ± 5 microsecond duration at the lower voltage level.

4.6.2.2 Operation initation, regulated output voltage, and internal reference.- Apply the pulse specified in 3.5.2 to Pin 1 of the assembly. With the frequency of the square wave input to Pin 15 set at 300 hertz, the voltage, sustained current, and interval reference shall be as specified in 3.5.4, and 3.5.5.

NOTE: The initial setting of internal reference was established in 4.6.2.1. No further adjustments to the internal reference shall be made once it has been set.

4.6.2.3 Voltage control.- Increase the frequency of the square wave generator toward 400 hertz, the voltage shall be as specified in 3.5.4 until the frequency reaches 400 ± 5 hertz, then the voltage shall be reduced to zero. The voltage shall not reappear until operation is again initiated by 3.5.2.

4.6.2.4 Pulse outputs.- Repeat the operations of 4.6.2.2 and 4.6.2.3. As the frequency reaches 400 ± 5 hertz, the pulses specified in 3.5.7 shall be observed.

4.6.2.5 Input current (operating state), input current (quiescent state).- Repeat the operations of 4.6.2.2 and 4.6.2.3. The current drawn by the assembly shall not exceed 350 milliamperes when the frequency of the square wave input is less than 395 hertz (I.E. when the regulated output voltage is being supplied to the load across Pins 3 and 7) and shall not

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exceed 10 milliamperes when the input square wave frequency exceeds 405 hertz.

4.6.2.6 Power off pulse.- Completely remove the positive side of the input voltage (Ei) from the assembly. As the voltage is removed, a pulse as specified in 3.5.8 shall be generated.

4.6.3 Electrical continuity.- A continuity test shall be made utilizing standard measuring equipment (SME) to determine conformance to the requirements of 3.6.

4.6.4 Interface requirements.- Perform this test using two test covers. One cover designed to the maximum limit of each interface with a tolerance of 10% of total limit, and one cover designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limit. Interfaces shall be provided as specified in 3.7 and as depicted on Drawings F10540966 and F10540501. Assembly shall be capable of being properly positioned and secured to both test covers. A test connector with interface conforming to requirements of Drawing C10540647 shall be assembled to circuit board as depicted on Drawing F10540501. Continuity shall be indicated between connector pins and conductors on circuit board. The Assembly Connector C10540938 shall be engaged to a test connector with interfaces conforming to requirements of Drawing C10541003. Continuity with test connector pin shall be indicated.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10541270. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The circuit controls the operation of the Q switch timing motor and triggers the flash lamp when the motor reaches the required speed for Q switching. The circuit is a module of the Range Finder AN/GVS-3.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Applicable packaging data sheet number (see 5.1)
c. Selection of applicable levels of preservation, packaging, and packing
d. Applicable stock number
e. Requirements for First Article testing

Custodian:

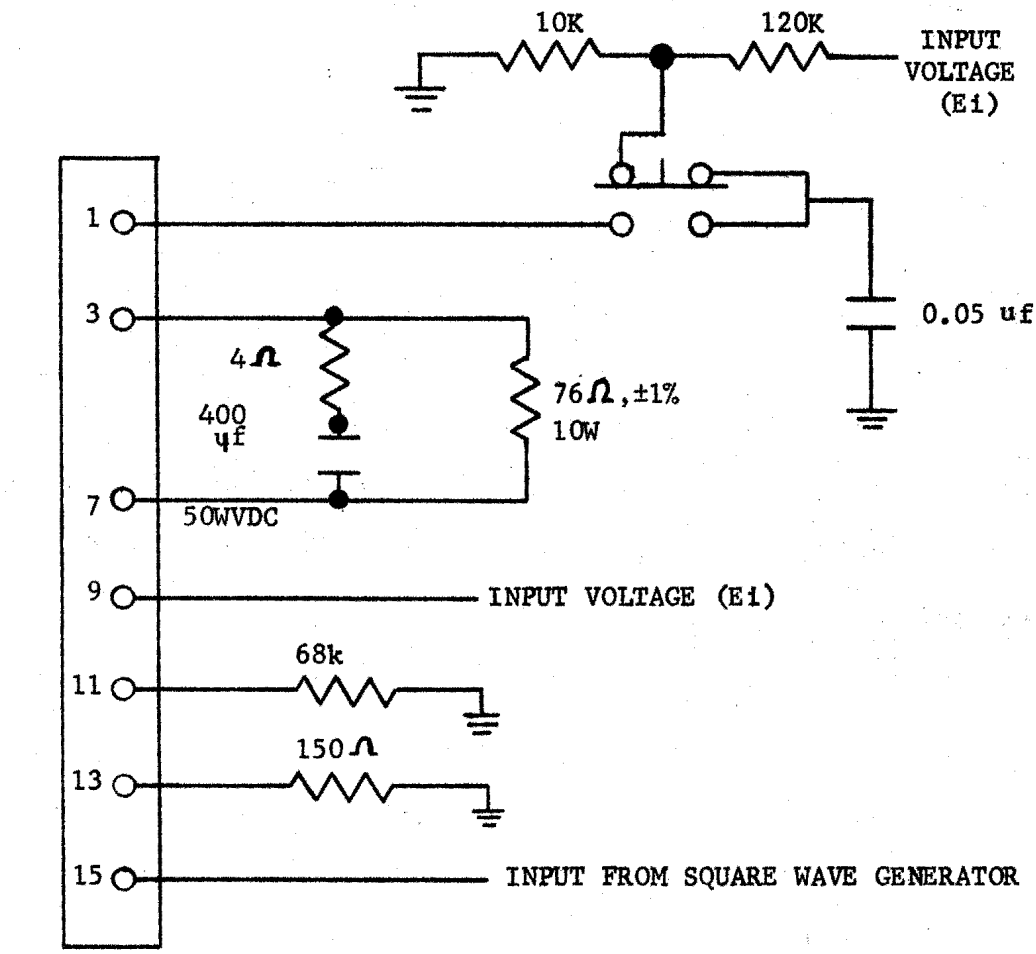
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Preparing activity:

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NOTE 1. ALL EVEN NUMBERED PINS ARE SYSTEM GROUND.

2. RESISTORS 1/4 WATT, RESISTANCE TOLERANCE ±5% UNLESS OTHERWISE SPECIFIED.

3. TOLERANCE ON CAPACITORS ±10%.

FIGURE I

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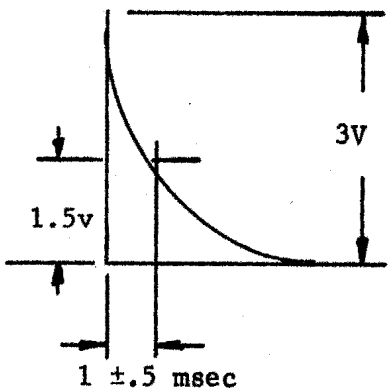


FIGURE 2

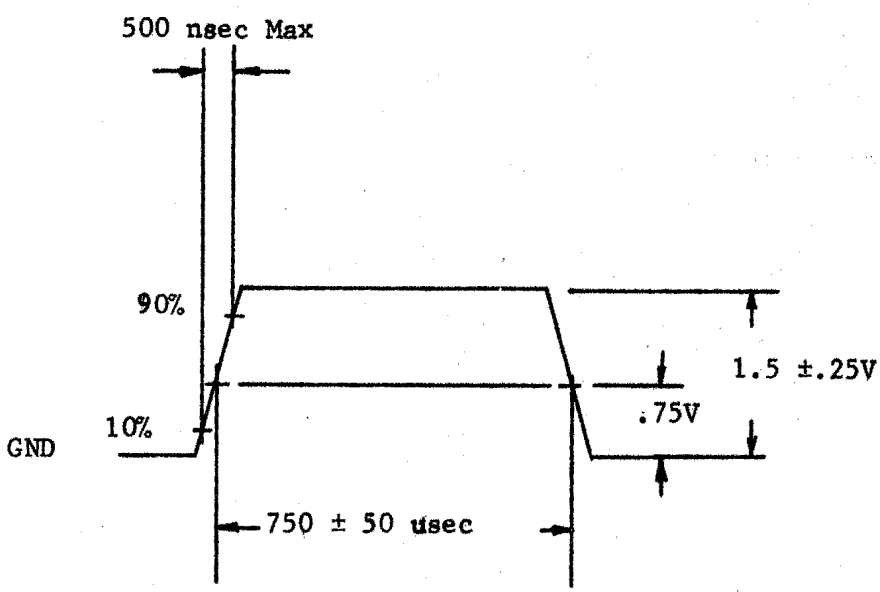


FIGURE 3

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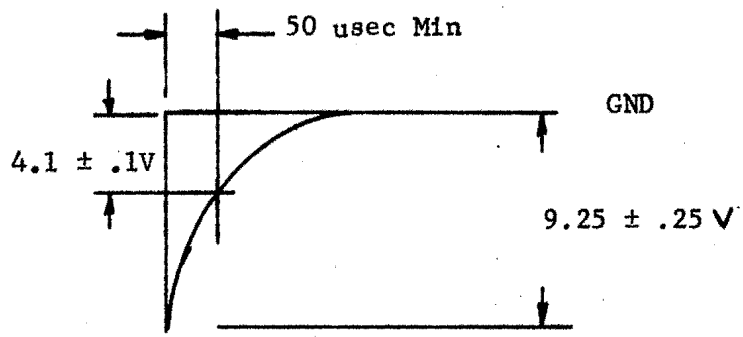


FIGURE 4

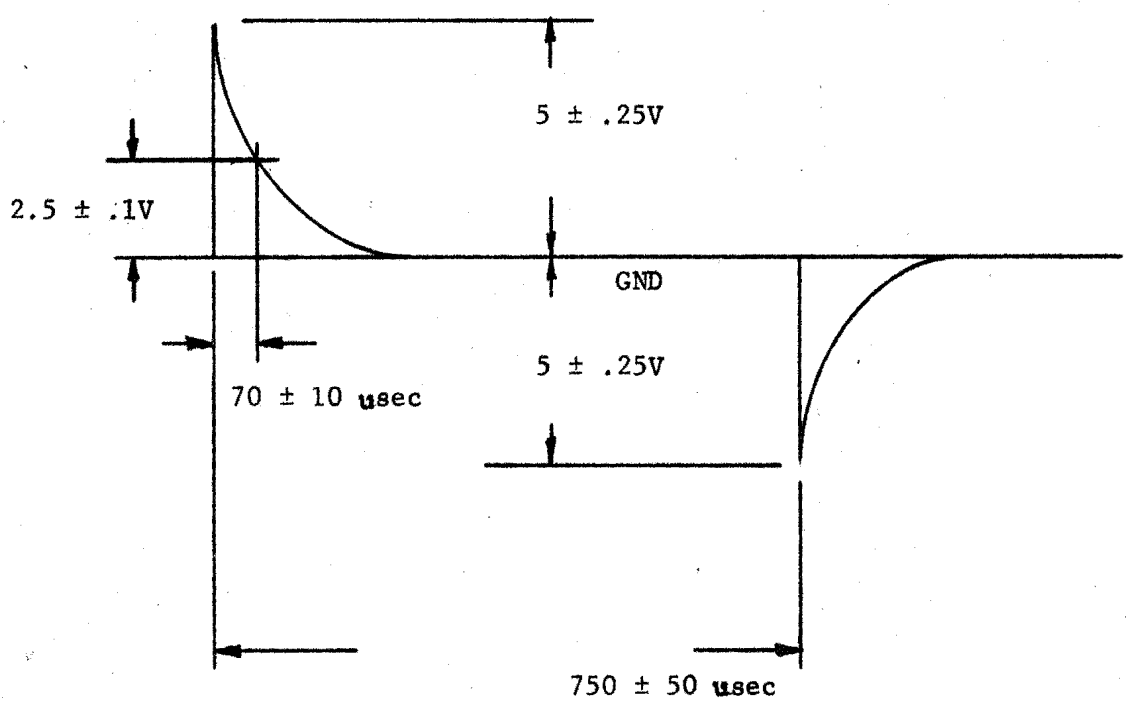


FIGURE 5

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